

THERMODYNAMICS

Tutorial – Week 1

Lecturer: Dr. Jay Wang

Teaching Assistant:

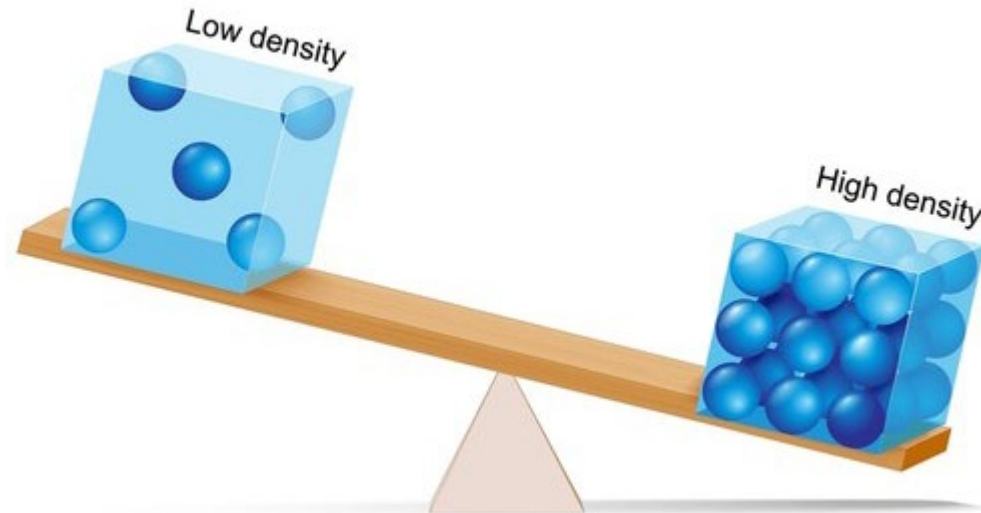
Mahir Ashef Hridoy & Cameron Gillingham



Density - ρ ($\frac{kg}{m^3}$)

Definition: density is a measure of how much mass (kg) can be stored in a given volume (m^3).

Formula: $\rho = \frac{m \text{ (mass)}}{V \text{ (volume)}}$

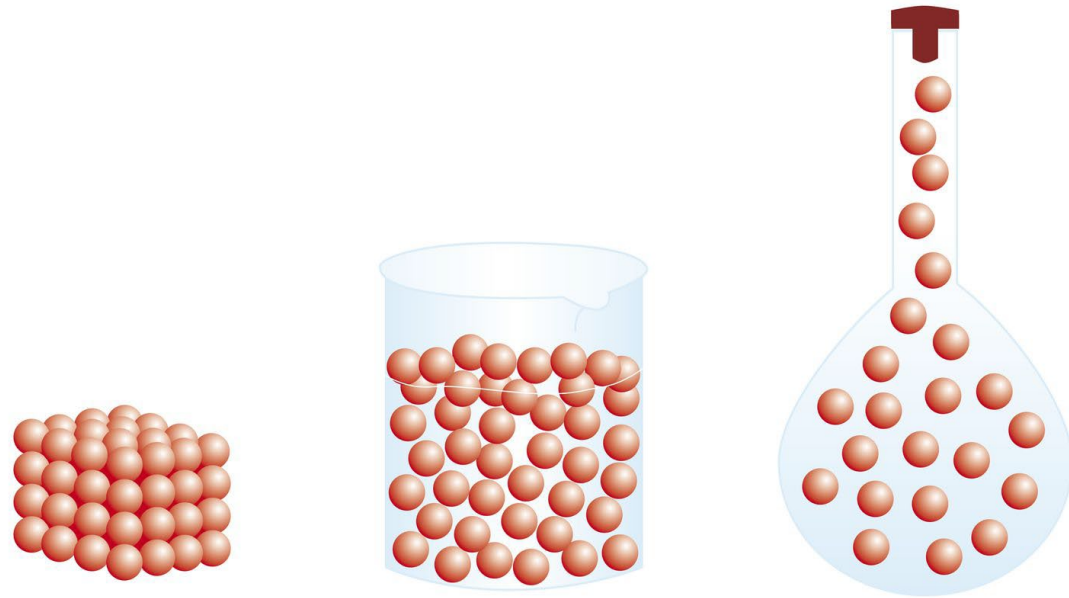




Specific Volume – $v \left(\frac{m^3}{kg} \right)$

Definition: specific volume is a measure of how much space/volume (m^3) is required to contain a given mass (kg).

Formula: $v = \frac{V \text{ (volume)}}{m \text{ (mass)}}$

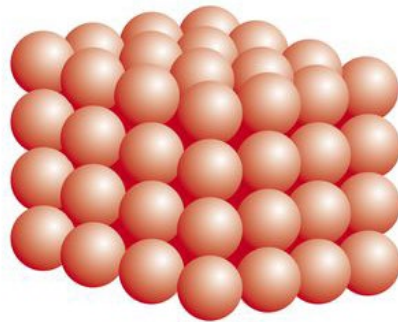




Specific Weight – γ ($\frac{N}{m^3}$)

Definition: specific weight is a measure of the weight (N) of a substance per unit volume (m^3).

$$\text{Formula: } \gamma = \frac{F_w \text{ (weight)}}{V \text{ (volume)}} = \frac{m \cdot g \text{ (mass} \times \text{gravity)}}{V \text{ (volume)}} = \rho \cdot g \text{ (density} \times \text{gravity)}$$



Q1. The mass of air in a room $3 \text{ m} \times 4 \text{ m} \times 25 \text{ m}$ is known to be 350 kg . Determine the density, specific volume, and specific weight.

Solution:

$$\rho = m/V = 350/(3 \times 4 \times 25) = \mathbf{1.167 \text{ kg/m}^3}$$

$$v = \rho^{-1} = \mathbf{0.857 \text{ m}^3/\text{kg}}$$

$$\gamma = \rho g = 1.167 \times 9.81 = \mathbf{11.45 \text{ N/m}^3}$$

Q2. Complete the following if $g = 9.81 \text{ m/s}^2$ and $V = 10^4 \text{ L}$.

Solution:

	$v(\text{m}^3/\text{kg})$	$\rho(\text{kg}/\text{m}^3)$	$\gamma(\text{N}/\text{m}^3)$	$m(\text{kg})$	$W(\text{N})$
(a)	20.00	0.05	0.49	0.50	4.91
(b)	0.50	2.00	19.62	20.00	196.20
(c)	2.45	0.41	4.00	4.08	40.00
(d)	0.10	10.00	98.10	100.00	981.00
(e)	0.98	1.02	10.00	10.19	100.00

Formulas: $\rho = v^{-1}$, $m = \rho V$, $W = mg$, $\gamma = \rho g$.

Q3. Determine the weight of a mass of 12 kg at a location where the acceleration of gravity is 9.77 m/s^2 .

Solution:

$$W = mg = 12 \text{ kg} \times 9.77 \text{ m/s}^2 = \mathbf{117.24 \text{ kg} \cdot \text{m/s}^2}$$

Q4. Calculate the force necessary to accelerate a 8000-kg rocket vertically upward at the rate of 28 m/s^2 .

Solution:

$$F_{\text{thrust}} = \text{Weight} + \text{Force for net upward acceleration} = mg + ma$$

$$F = 8000 \times 9.81 + 8000 \times 28 = 78,480 + 224,000 = \mathbf{302,480 \text{ N}}$$

Q5. A 2200-kg automobile traveling at 90 kph hits the rear of a stationary, 1000-kg automobile. After the collision, the large automobile slows to 60 kph, and the smaller vehicle has a speed of 78 kph. What has been the increase in internal energy, taking both vehicles as the system?

Solution:

Before collision: $V_{a1} = 90\text{kph} = \frac{90 \times 10^3 \text{m}}{3600\text{s}} = 25 \text{ m/s}$, thus: $E_{k1} = \frac{1}{2} m_{a1} V_{a1}^2 = 687,550 \text{ J}$

After the collision: $V_{a2} = 60\text{kph} = 16.67\text{m/s}$ and $V_b = 78\text{kph} = 21.67 \text{ m/s}$, the kinetic energy is: $E_{k2} = \frac{1}{2} m_{a2} V_{a2}^2 + \frac{1}{2} m_b V_b^2 = 540,472 \text{ J}$

Based on conservation of energy: $E_{k1} + U_1 = E_{k2} + U_2$

Thus: $U_2 - U_1 = E_{k1} - E_{k2} = 687,550 - 540,472 = \mathbf{147,078 \text{ J}}$

Q6. Calculate the force due to the pressure acting on the 1.2-m diameter horizontal hatch of a submarine submerged 800 m below the surface.

Solution:

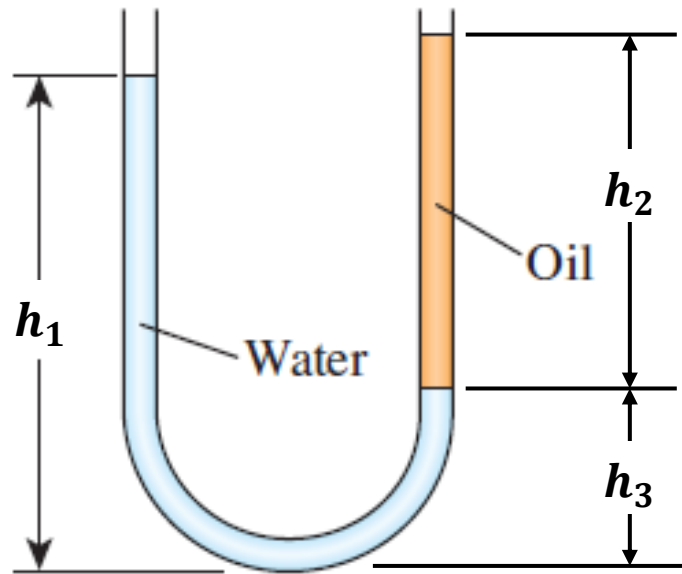
$$p = \rho gh = (1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(800 \text{ m}) = 7.848 \times 10^6 \text{ kg/(m} \cdot \text{s}^2) = 7.848 \text{ MPa}$$

$$F = pA = (7.85 \times 10^6 \text{ N/m}^2) \left[\frac{\pi(1.2^2)}{4} \text{ m}^2 \right] = 8.88 \times 10^6 \text{ N}$$

Q7. A U-tube with arms open to the atmosphere. Water is poured into the U-tube from one arm, and light oil from the other. One arm contains water, while the other arm contains both fluids. Determine the height of each fluid in that arm.

Given: $\rho_{\text{oil}} = 790 \text{ kg/m}^3$, $h_1 = 50\text{cm}$ and $h_3 = \frac{1}{4}h_2$.

Solution:



$$p_{\text{left}} = p_{\text{right}} \Rightarrow \rho_{\text{water}} \cdot g \cdot (h_1 - h_3) = \rho_{\text{oil}} \cdot g \cdot h_2.$$

$$\text{Given } h_1 = 50\text{cm} = 0.5\text{m} \text{ and } h_3 = \frac{1}{4}h_2.$$

$$\Rightarrow 1000 \times \left(0.5 - \frac{1}{4}h_2\right) = 790 \times h_2$$

$$\Rightarrow 500 - 250 \cdot h_2 = 790 \cdot h_2$$

$$h_2 = \frac{500}{1040} = \mathbf{0.48 \text{ m}} \text{ and } h_3 = \frac{1}{4}h_2 = \mathbf{0.12 \text{ m}}$$

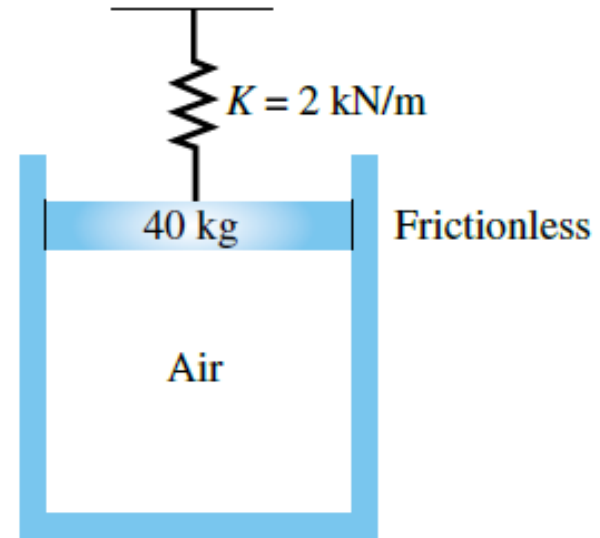
Q8. Calculate the pressure in the 150-mm diameter cylinder. The spring is compressed 35 cm.

Solution:

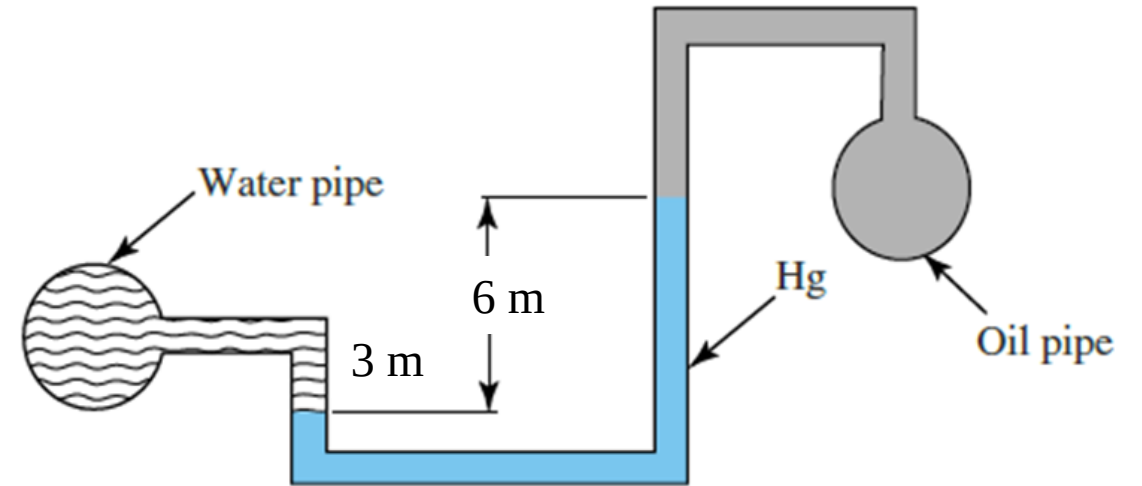
Force balance:

$$p_{\text{air}} \cdot A = m_{\text{piston}} \cdot g + K \cdot x$$

$$p_{\text{air}} = \frac{40 \times 9.81 + 2000 \times 0.35}{\pi \left(\frac{0.15}{2} \right)^2} = \frac{392.4 + 700}{0.0177} = \mathbf{61717.51 \text{ Pa}}$$



Q9. Determine the pressure difference between the water pipe and the oil pipe. The top of the mercury column is at the same elevation as the middle of the oil pipe.



Solution:

$$\Delta p = p_{\text{water}} - p_{\text{oil}} = \rho_{\text{Hg}} \cdot g \cdot h_{\text{Hg}} - \rho_{\text{water}} \cdot g \cdot h_{\text{water}}$$

$$\Delta p = 9.81 \times (13600 \times 6 - 1000 \times 3) = 9.81 \times 78600 = \mathbf{771,066 \text{ Pa}}$$

Q10. A horizontal 2.5-m-diameter gate is located in the bottom of a water tank. Determine the force F required to just open the gate.

Solution:

To open the gate, the moment created by the applied force must be at least equal to the moment due to water pressure. The hydrostatic force acts at the centre of the gate at a distance of $R = \frac{D}{2} = 1.25\text{m}$ from the hinge.

Thus, $F \cdot D = F_{\text{water}} \cdot \frac{D}{2}$, then $F = \frac{F_{\text{water}}}{2}$.

Pressure at the gate: $p = \rho gh = 1000 \frac{\text{kg}}{\text{m}^3} \times 9.81 \frac{\text{m}}{\text{s}^2} \times 6\text{m} = 58,860 \frac{\text{kg}}{\text{m} \cdot \text{s}^2} = 58,860 \text{ Pa}$

$F_{\text{water}} = pA = 58860 \times \pi \left(\frac{2.5\text{m}}{2}\right)^2 = 288928.35\text{N}$. Then $F = \frac{288928.35}{2} = \mathbf{144464.17 \text{ N}}$

